

The Structure of a LysM Domain from *E. coli* Membrane-bound Lytic Murein Transglycosylase D (MltD)

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The LysM domain is a widespread protein module. It was originally identified in enzymes that degrade bacterial cell walls but is also present in many other bacterial proteins. Several proteins that contain the domain, such as Staphylococcal IgG binding proteins and *Escherichia coli* intimin, are involved in bacterial pathogenesis. LysM domains are also found in some eukaryotic proteins, apparently as a result of horizontal gene transfer from bacteria. The available evidence suggests that the LysM domain is a general peptidoglycan-binding module. We have determined the structure of this domain from *E. coli* membrane-bound lytic murein transglycosylase D. The LysM domain has a $\beta\alpha\beta$ secondary structure with the two helices packing onto the same side of an anti-parallel β sheet. The structure shows no similarity to other bacterial cell surface domains. A potential binding site in a shallow groove on surface of the protein has been identified.

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Introduction

Peptidoglycan is a component of the cell walls of both Gram positive and Gram negative bacteria, where it provides mechanical support and prevents bacteria bursting under high internal osmotic pressure. It is composed of linear chains of *N*-acetylmuramic acid and *N*-acetylglucosamine, cross-linked by short peptides. The importance of peptidoglycan is highlighted by the fact that many antibiotics block its biosynthesis. Bacteria and bacteriophage produce a variety of enzymes that cleave peptidoglycan either to lyse host cells or to re-model the cell wall during growth or cell division. Different classes of peptidoglycan degrading enzymes cleave the molecule at different points. Some, such as muramidases and lysozymes, attack the glycosidic linkages of the glycan strand. Amidases cleave the amide bond between *N*-acetyl muramic acid and the L alanine that forms part of the peptide cross-links, and peptidases cut amide bonds in the peptide cross-links. Many of these enzymes are modular and are composed

of catalytic units linked to a number of other domains. The extracellular muramidase 2 of *Enterococcus hirae*, for example, consists of a catalytic N-terminal domain fused to a C-terminal region containing multiple repeats of approximately 40 amino acid residues (Joris *et al.*, 1992). This sequence motif is found in many other enzymes involved in cell wall degradation and is also present in a number of other proteins, many of which are known to be associated with bacterial cell walls (Birkeland, 1994) (Figure 1). It appears that this motif codes for a protein domain that probably has a general peptidoglycan binding function. Since it was originally identified in bacterial lysins, the repeats have been termed LysM domains for lysin motif (Ponting *et al.*, 1999). To understand further the structure and function of this module we have determined the structure of a LysM domain from *Escherichia coli* membrane-bound lytic murein transglycosylase D (MltD). MltD consists of an N-terminal transglycosylase domain with two LysM repeats at the C terminus (Figure 1). The exact function of MltD is unknown. However, based on the phylogenetic distribution of the protein it has recently been suggested that it may be involved in flagellar biosynthesis (Pellegrini *et al.*, 1999).

Abbreviations used: MltD, membrane-bound lytic murein transglycosylase D.

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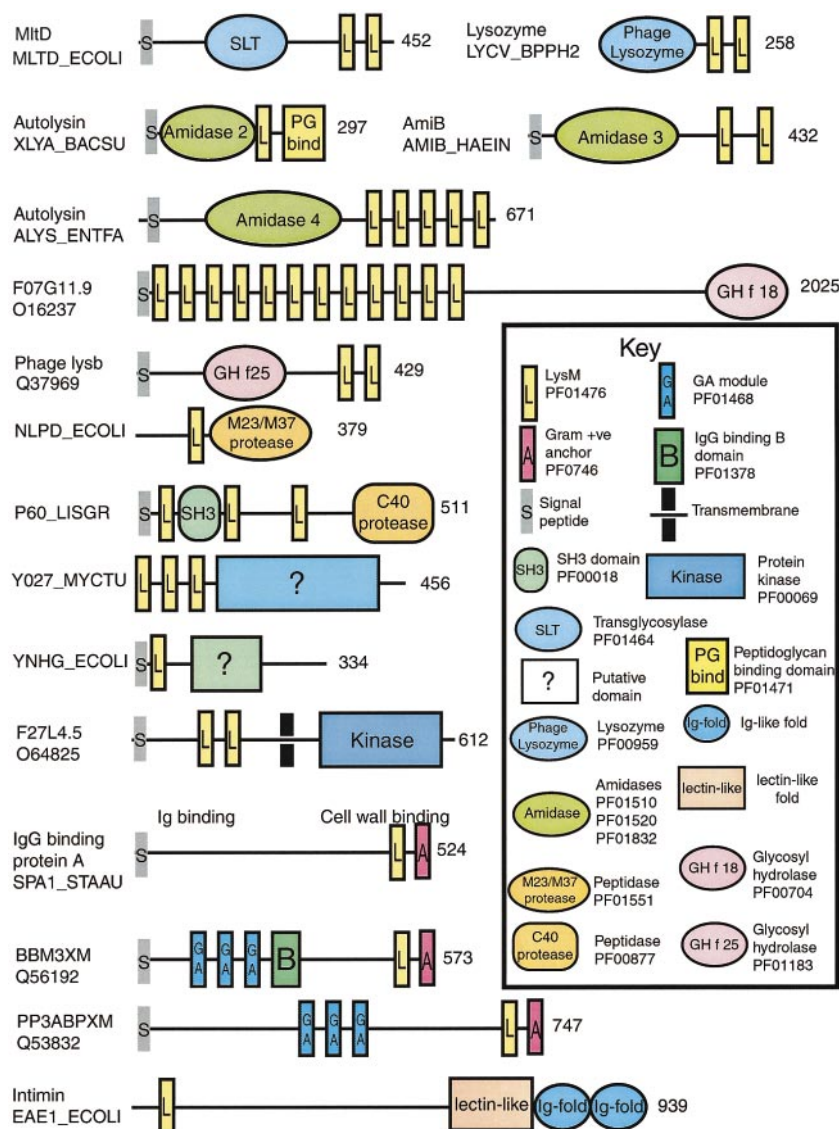


Figure 1. Schematic representation of the domain architectures of representative proteins containing the LysM domain. The Pfam (Bateman *et al.*, 1999) accession number of each domain is given.

Results

Residues 389 to 452 of *E. coli* MltD encompassing the C-terminal copy of the two motifs in this enzyme, were overexpressed in *E. coli* and purified to homogeneity. Assignments for this peptide were obtained using a range of 2D and 3D heteronuclear NMR experiments using isotopically enriched samples. Non-random coil chemical shifts and long-range NOEs were only observed for the residues between 398 and 445, and structures were calculated for this region. Residues will now be referred to with Asp398 as residue 1. A total of 100 structures were calculated and the 20 structures with the lowest number of constraint violations were selected for further analysis (Figure 2(a)). None of these structures had distance constraint violations greater than 0.3 Å or angle constraint violations greater than 5°.

The LysM domain has a $\beta\alpha\beta$ secondary structure with the two helices packing onto the same

side of a two-stranded anti-parallel β sheet (Figure 2(b)). Searching the protein structure database using the program DALI (Holm & Sander, 1995) and the CATH database (Orengo *et al.*, 1997) revealed no significant structural similarities to any other domains.

A structure-based alignment of representative LysM domain sequences is given in Figure 3. The structured region extends only two to three residues at either end from the region of sequence similarity. Insertions and deletions occur only between elements of secondary structure. The largest insertion, of 22 residues, is in the loop between the two helices. The first part of the loop between the second helix and the second strand also varies in length. The latter part of the loop after L37 is, however, better conserved. The conserved hydrophobic residues all form part of a continuous hydrophobic core and are mostly involved in the packing of the two helices onto the sheet. All of these residues are located in the elements of

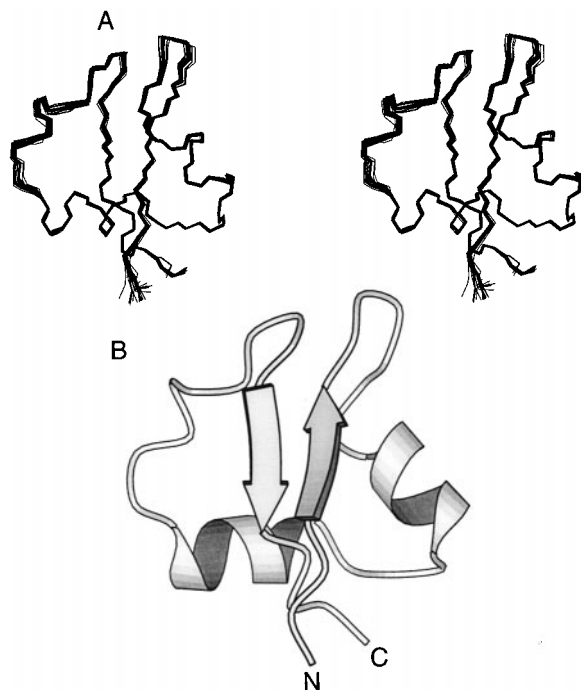


Figure 2. (a) Stereo view of the best fit superimposition of the backbone atoms of the 20 lowest energy NMR structures of the LysM domain. (b) Molscript (Kraulis, 1991) representation of the LysM domain.

secondary structure except for L37, which is in the loop between helix 2 and strand 2, and anchors this loop into the core. Residues W30 and F46 form a second, less well conserved, hydrophobic core that packs the end of helix 2 onto the side of the sheet opposite the main hydrophobic core.

One of the features of this sequence motif is the presence of two highly conserved glycine residues, corresponding to G10 and G40 in this structure. G10 is part of a tight turn at the end of strand 1 and G40 is located in the loop between helix 2 and sheet 2 where the chain turns sharply into the final strand of the sheet. An asparagine residue corresponding to N31 in this domain is also highly conserved. This residue appears to play a role in forming the turn at the end of helix 2. Residue 12 is a serine or threonine in most LysM domains, and acts as an N-cap for helix 1. Residue 11 is an aspartate or glutamate residue in most copies of this domain. This residue is situated in the turn between strand 1 and helix 1. From a structural point of view, it is not clear why this residue should always be acidic, it could be that it has a functional role.

Discussion

Here, we have demonstrated that the lysin motif codes for a folded globular domain. A number of other domains have been characterised from bacterial cell surface proteins and it is becoming clear

that these proteins, like many eukaryotic cell surface proteins, are highly modular in nature (Goward *et al.*, 1993). A variety of enzymes involved in bacterial cell wall degradation, for example, contain a 60-amino acid residue domain distinct from the LysM domain that is also believed to have a general peptidoglycan binding function. The structure of this domain is known (Dideberg *et al.*, 1982). It consists of three α -helices arranged in a bundle and is therefore structurally very different from the LysM domain. The LysM fold also shows no similarity to the structures of other small bacterial cell surface domains such as the B1 Ig-binding domain, that is composed of a single helix that packs against a mixed β -sheet (Gallagher *et al.*, 1994), or the GA module, that is a three-helix bundle (Johansson *et al.*, 1997).

The LysM domain is one of the most common modules in bacterial cell surface proteins. Searching the current sequence databases for copies of this domain reveals that it is present in more than 100 proteins. It occurs most often in cell wall degrading enzymes where it probably acts to anchor the catalytic domains to their substrates but is also present in many other bacterial proteins (Figure 1). A number of these proteins are involved in bacterial pathogenesis. Staphylococci produce IgG binding proteins that help the bacteria to avoid the host immune response (Uhlen *et al.*, 1984). The IgG binding region is located at the N terminus of the molecule. The C-terminal part of the protein binds to the bacterial cell wall (Goward *et al.*, 1993) and contains a LysM domain. Intimin is a 930-residue outer membrane protein required for intimate attachment of enterohaemorrhagic and enteropathogenic *E. coli* to mammalian cells (Jerse *et al.*, 1990). The C-terminal 280 amino acid residues of the protein mediate the bacterial host-cell interaction and contain two Ig-like folds and a C-type lectin-like domain (Frankel *et al.*, 1996; Kelly *et al.*, 1999). The N-terminal part of the protein is the most highly conserved region and contains a single copy of the LysM domain, suggesting that this portion of the molecule may interact with the bacterial cell wall. A LysM domain is also present at the N terminus of a *Streptococcus pyogenes* protein that may have a role in pathogenesis (Kil *et al.*, 1994).

LysM domains are not limited to bacteria but are also present in a number of eukaryotic proteins (Ponting *et al.*, 1999). Multiple copies of the domain are found in chitinases from the nematode *Caenorhabditis elegans* and the fungus *Kluyveromyces lactis*. The domains in these proteins contain four conserved cysteine residues that probably form disulphide bridges. The equivalent residues in the MltD domain are Asp1, Leu13, Thr34 and Thr44. Based on the structure it would appear that cysteines 1 and 4 and 2 and 3 are bonded (Figure 4). This would link the start of the domain to the end of the second strand of the sheet and the turn between strand 1 and helix 1 to the loop between helix 2 and strand 2. This conclusion is

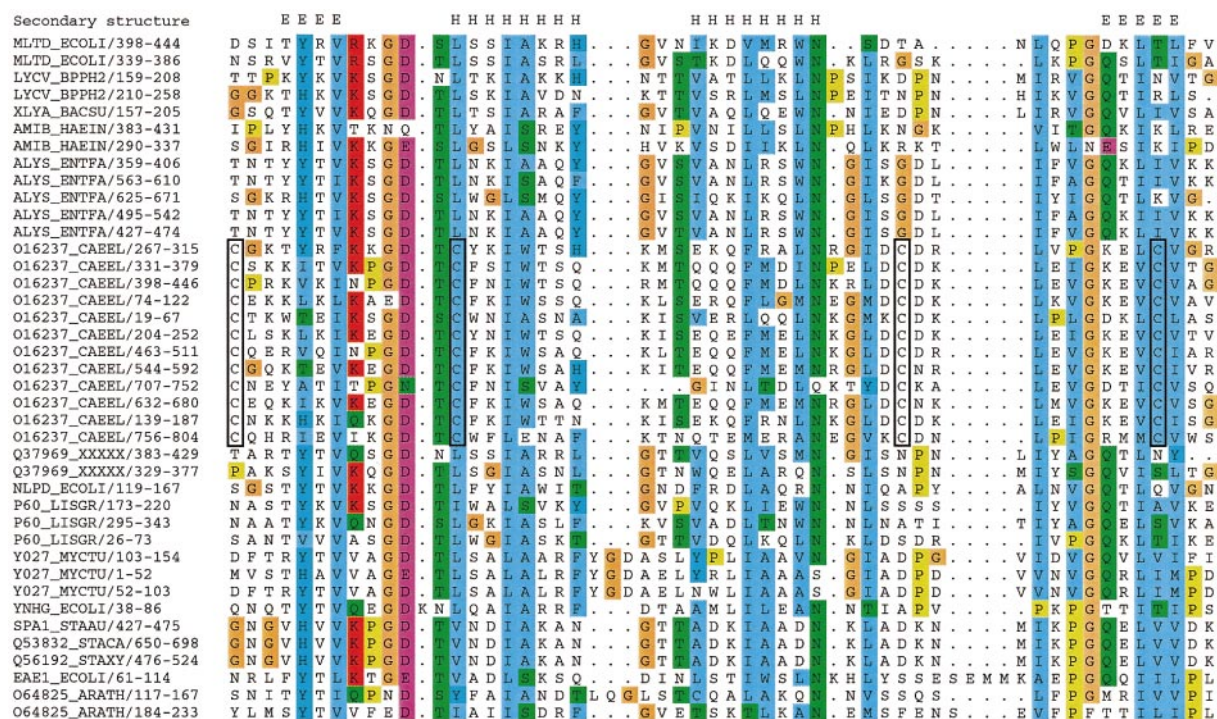


Figure 3. Structure-based alignment of representative sequences of LysM domains.

supported by the observation that a domain in a lipoprotein from *Aquifex aeolicus* has cysteine residues 1 and 4 but not 2 and 3. The disulphide bridges are clearly not essential for this fold. Several other examples are known where disulphide bridges have been introduced into extracellular eukaryotic homologues of prokaryotic domains (Doolittle, 1995) (Bork *et al.*, 1996). The disulphide bonds are believed to increase stability and resistance to proteolysis in these domains. One of the disulfide bridges may, however, have an additional role in the LysM domain. Many domains that are present in a tandem array such as Ig domains or fibronectin type III domains have the N and C termini at opposite ends of the module to facilitate head-to-tail linkage. For these domains, the linkers between modules are generally quite short. The N and C termini in the LysM domain, however, are close together. This would make it difficult to arrange the domain into a large array without the polypeptide chain changing direction between each copy of the domain. The linkers between LysM domains in tandem arrays are normally more than ten residues long, which would allow for this to occur. The largest arrays of LysM domains are found in chitinases in which the domains contain the disulphide bridges discussed above. From the structure it appears that the polypeptide would have to fold back on itself to allow cysteine residues 1 and 4 to bond (Figure 4) and this disulphide bridge may have a role in helping to arrange the domains in large tandem arrays.

The LysM domain is also present as a single copy in several uncharacterised *C. elegans* proteins

where the cysteine residues are not conserved. It is also found in a number of *Arabidopsis thaliana* proteins, including one where it is N-terminal to a kinase domain, suggesting a role in signal transduction. A chitinase from the plant *Volvox carteri* and a glycoprotein from the fungus *Colletotrichum lindemuthianum* also contain LysM domains. A LysM domain is found in some isoforms of the *Drosophila* L82 gene product, a protein involved in metamorphosis (Stowers *et al.*, 1999) and in an uncharacterised *Drosophila* protein.

The distribution of LysM domains in eukaryotes is patchy. It is most common in organisms that are closely associated with bacteria. It is not present in any archaeal proteins or in any proteins from *Saccharomyces cerevisiae*, and no copies have so far been detected in any human proteins. It seems most likely that the presence of this domain in eukaryotic proteins is the result of horizontal gene transfer from bacteria. It is possible that some of these proteins have an anti-bacterial function and the LysM domain has been recruited to bind bacterial cell walls.

All the available data suggests that the LysM domain has a peptidoglycan-binding role (Joris *et al.*, 1992), although the precise component to which it binds has not been established. As discussed above, the conserved residue that is most likely to be involved in an interaction with a ligand is Asp11. This is located in the loop between strand 1 and helix 1. Another residue in this loop corresponding to Arg12 in the MltD domain is conserved in many LysM domains. Interestingly, this loop varies the least in size. Currently only one

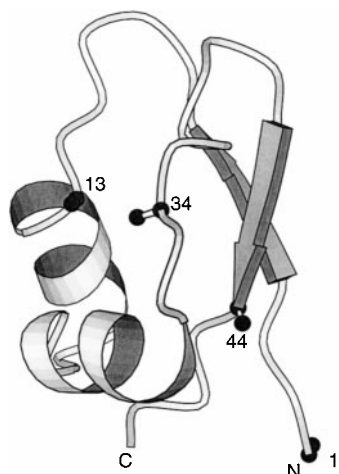


Figure 4. Molscript (Kraulis, 1991) representation of the LysM domain showing the positions of the C $^{\alpha}$ and C $^{\beta}$ of the amino acid residues equivalent the cysteine residues in the LysM domains of *C. elegans* and *Kluyveromyces lactis* chitinases.

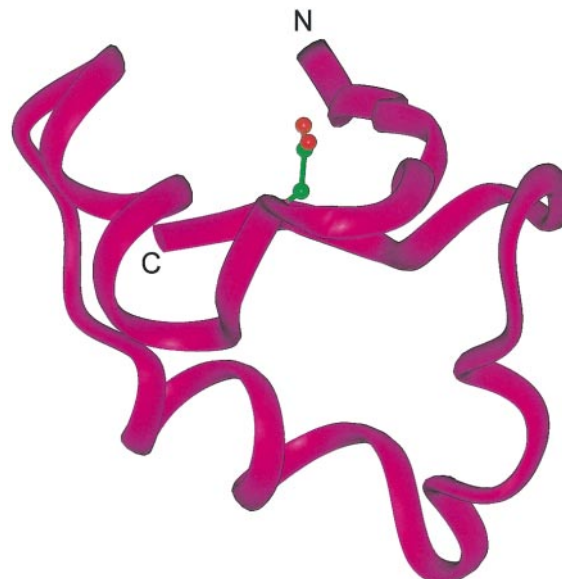


Figure 5. Structure of the LysM domain showing the position of Asp11 in the putative ligand binding site.

protein is known that has an insertion in this loop and this only of one residue. The loop between strand 1 and helix 1 lies at the end of a shallow groove on the surface of the domain and this region is the best candidate for a ligand-binding site (Figure 5).

There is considerable evidence to indicate that protein-peptidoglycan interactions play an important role in the organisation of the bacterial cell envelope. The wide distribution of the LysM domain in bacterial proteins especially those with a role in pathogenesis make it a potential target for agents that block its function. The results reported here can be used as the basis for further work to identify ligands for the domain and to determine the basis of the interaction.

Materials and Methods

Sample preparation

The DNA sequence coding for residues 389 to 452 of *E. coli* MltD was amplified from *E. coli* chromosomal DNA using PCR and cloned into the pRSETa expression vector (Invitrogen). The LysM domain was expressed in C41 (DE3) cells (Miroux & Walker, 1996) grown in M9 media with appropriately labelled carbon and nitrogen sources. Protein was purified by ion exchange chromatography using DEAE fast flow Sepharose (Pharmacia) followed by gel filtration on a G75 Superdex column (Pharmacia). NMR samples were 3 mM protein in 20 mM phosphate buffer (pH 5.8).

Structure determination

NMR spectra were recorded at 25 °C using a Bruker AMX 500 spectrometer equipped with a pulsed field gradient triple resonance probe. Resonance assignments for

the domain were obtained using standard double and triple resonance methods using isotopically enriched protein samples reviewed (Bax & Grzesiek, 1993). Briefly, backbone assignments were obtained using 3D ^1H ^{15}N edited TOCSY, HBHACONH, CBCACONH and HNCACB spectra. Additional side-chain assignments were obtained using an HCCH-TOCSY experiment. The assignments have been deposited in the BioMagResBank accession number 4680. Distance restraints were obtained from the analysis of ^1H ^1H 2D NOESY, 3D ^1H ^{15}N edited NOESY and 3D ^1H ^{13}C edited NOESY experiments, all recorded with a 100 ms mixing time. Peak

Table 1. Structural statistics

Root mean squared deviations from experimental restraints	
Distance (Å)	0.043 $\pm$ 0.008
Angle (deg.)	0.70 $\pm$ 0.06
Root mean squared deviations from idealized covalent geometry	
Bonds (Å)	0.003793 $\pm$ 4.281e-05
Angles (deg.)	0.55 $\pm$ 0.006
Impropers (deg.)	0.52 $\pm$ 0.008
Root mean squared distribution around the mean coordinate positions	
All backbone atoms (Å)	0.26 $\pm$ 0.08
All heavy atoms (Å)	0.92 $\pm$ 0.10
Measures of structural quality determined using PROCHECK (Laskowski <i>et al.</i> , 1996) Ramachandran plot	
% residues in most favoured regions	86.54 $\pm$ 1.78
% residues in additionally allowed regions	13.46 $\pm$ 1.78
% residues in generously allowed regions	0.0
% residues in disallowed regions	0.0
All statistics are quoted $\pm$ one standard deviation.	

volumes were converted into inter-proton distance restraints based on known distances. The distance restraints were then classified as strong, medium or weak corresponding to upper distance bounds of 2.7 Å, 3.3 Å, and 5.0 Å. An additional 0.5 Å was added to upper distance bounds for atoms involving methyl group protons. A lower distance bound of 1.8 Å was used for all NOE-derived distance restraints. The program Aqua (Laskowski *et al.*, 1996) was used to remove redundant distance constraints. Structures were calculated using 99 intra-residue, 136 sequential 106 medium range and 152 long-range NOE restraints. Some 15 hydrogen bond restraints were also used in the calculations. Potential hydrogen bond donors were assigned from a ^1H ^{15}N HSQC spectrum recorded immediately after dissolving a protein sample in 100% $^1\text{H}_2\text{O}$. Hydrogen bonds were then identified based on the characteristic NOE patterns that are observed for residues in regular secondary structure, together with the solvent exchange data (Wüthrich, 1986). Torsional restraints for 27 ϕ and 20 ψ angles were obtained from an analysis of C' N, C^α H^α and C^β chemical shifts using the program TALOS (Cornilescu *et al.*, 1999). Chi 1 side-chain restraints for 14 residues were determined from $\text{J}\alpha\beta$ coupling constants and NOE data. Stereospecific assignments for the methyl groups of valine and leucine residues were obtained from the splitting patterns in a high-resolution ^1H ^{13}C HSQC spectrum of a 10% fractionally ^{13}C -labelled sample (Neri *et al.*, 1989). Structures were calculated using simulated annealing from random starting structures followed by restrained minimisation using the program XPLOR 3.8 (Brünger, 1992).

Accession numbers

The coordinates have been deposited in the protein structure database, code 1EoG.

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